

Summary

Research / Applied Scientist with a Ph.D. in Biostatistics from Duke University, specializing in causal inference, experimentation, and large-scale advertising ML. Currently at Upstart, driving methodology across uplift modeling, offline evaluation, online exploration, calibration, and forecasting to optimize decisions under fixed channel cadences, long-term feedback loops, and rare outcomes in dynamic environments.

Professional Experience

Upstart

Senior Research Scientist
Research Scientist

Seattle, WA

Dec 2025 – Present

Jun 2025 – Dec 2025

- Develop targeted selection logic, model overlap frameworks, and surrogate models to optimize experimentation budgets. Through strategic allocation, these frameworks achieve a 70% reduction in Minimum Detectable Effect (MDE) for rare outcomes alongside a 3% boost in campaign performance.
- Design dynamic calibration for send optimization and build exploration-based frameworks for off-policy offline evaluation, navigating the high costs of failure and strict physical constraints of the channel.
- Partner closely with Growth Marketing, Direct Mail, and Product teams to identify strategic opportunities for model optimization, aligning causal research with key business goals and channel constraints.
- Lead the migration of our Direct Mail uplift modeling codebase to a modern Metaflow and Python pipeline, achieving a 20x speedup in execution time and significantly reducing compute costs.

Meta

Research Scientist Intern

Seattle, WA

May 2024 – Aug 2024

- Developed a domain-adapted, model-based approach for offline counterfactual evaluation of large-scale ad-ranking models, improving pre-deployment testing of ML systems. Work accepted at the 2024 *Causality, Counterfactuals & Sequential Decision-Making Workshop* at RecSys and the 2024 *Conference on Digital Experimentation @ MIT*.
- Designed and ran empirical analyses comparing multiple evaluation methods, identifying conditions where the proposed approach outperforms existing benchmarks.
- Bridged observational causal inference and experimentation by enhancing offline evaluation techniques for models later tested via A/B experiments.

Duke University

Ph.D. Researcher

Durham, NC

Aug 2021 – May 2025

- Formulated a novel partial identification and data fusion framework using double machine learning to reliably estimate causal effects from both experimental and observational data under simultaneous assumption violations. Work published at NeurIPS.
- Developed an interpretable method for estimating dynamic treatment regimes that reduced the probability of an adverse medical event by over 20 percentage points when operationalized on a real-world ICU dataset. Work published at AISTATS.
- Created a variant of multi-label learning optimized for rare event settings that improved prediction of cardiovascular adverse events in pregnancy by up to 10%. Work published as a spotlight paper at UAI.
- Engineered a new method for large scale, computationally inexpensive, and interpretable causal inference that is over 100 times faster than existing benchmarks. Work published at UAI.

Optum

Data Scientist

Eden Prairie, MN

Jun 2019 – Jul 2021

- Built and deployed a multimodal time-series model assessing the risk of every development team's proposed application update.
- Detected suspicious activity as lead python programmer for fraud investigation in the payout of the U.S. Government's COVID-19 HHS CARES Act Provider Relief Fund facilitated by Optum.

- Automated development team's Jenkins environment by creating customizable Jenkinsfiles to deploy machine learning models to Kubernetes.
- Summarized bank transaction and account activity logs into feature vectors using PySpark, and used Spark's MLlib library to detect irregular account activity.

Education

Duke University

Ph.D. in Biostatistics & Bioinformatics

Dissertation: *Leveraging Machine Learning for Interpretable Causal Inference*

Advisors: Cynthia Rudin, David Page, & Alexander Volfovsky

Durham, NC

Aug 2021 – May 2025

Loyola Marymount University

B.S. in Applied Mathematics (Minor in Biochemistry)

Summa Cum Laude

Thesis: *Neural Machine Translation & Deep Learning*

Advisor: Thomas Laurent

Los Angeles, CA

Aug 2015 – May 2019

Publications

Quinn Lanners, Cynthia Rudin, Alexander Volfovsky, and Harsh Parikh. Data fusion for partial identification of causal effects. *Advances in neural information processing systems*, 38:150638–150702, 2026

Quinn Lanners*, Qin Weng*, Marie-Louise Meng, and Matthew M Engelhard. Common event tethering to improve prediction of rare clinical events. In *The 40th Conference on Uncertainty in Artificial Intelligence*, 2024

Harsh Parikh*, **Quinn Lanners***, Zade Akas, Sahar Zafar, M Brandon Westover, Cynthia Rudin, and Alexander Volfovsky. Safe and interpretable estimation of optimal treatment regimes. In *International Conference on Artificial Intelligence and Statistics*, pages 2134–2142. PMLR, 2024

Quinn Lanners, Harsh Parikh, Alexander Volfovsky, Cynthia Rudin, and David Page. Variable importance matching for causal inference. In *Uncertainty in Artificial Intelligence*, pages 1174–1184. PMLR, 2023

Mohamed A Radwan, Himaghna Bhattacharjee, **Quinn Lanners**, Jiasheng Zhang, Serkan Karakulak, Houssam Nas-sif, and Murat Ali Bayir. Counterfactual evaluation of ads ranking models through domain adaptation. In *Causality, Counterfactuals & Sequential Decision-Making workshop at RecSys*. arXiv:2409.19824, 2024

* Denotes co-first authorship.

Technical Skills

Languages: Python, SQL

ML & Frameworks: XGBoost, PyTorch, Scikit-learn, EconML, CausalML, PySpark, d3rlpy, Pandas, NumPy

Infrastructure & Tools: Metaflow, Databricks, Redshift, AWS, Docker, Kubernetes, Jenkins, Slurm, Git

Methodologies: Causal Inference, Uplift Modeling, Off-Policy Evaluation, Contextual Bandits, CUPED, Synthetic Controls, Instrumental Variables, Online Experimentation, Surrogate Modeling, Partial Identification